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Big five personality and COVID-19 beliefs, behaviors, and vaccine intentions: The mediating role of political ideology

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Abstract

Because personal health decisions can impact the health of the broader community, researchers have increasingly sought to understand the psychological bases for different responses to public health communications and prescriptions. We contribute to this literature in two ways. First, we analyze the relationship between Big Five personality traits and three critical beliefs and behaviors in the context of the COVID-19 pandemic. These are vaccine hesitancy, engagement in risky (vs. self-protective) pandemic social behaviors, and conspiratorial beliefs surrounding the origins, consequences, and public health response to COVID-19. Second, we draw on theory from political psychology to model the joint effects of personality and ideology. Our analysis of two American samples (MTurk = 510; Representative = 441) indicated that political liberalism mediated the relationship between Openness and COVID-related attitudes and behaviors.

KEYWORDS
health, health psychology, personality, political psychology, social psychology and personality

1 | INTRODUCTION

The question of if or how individual differences shape important social behaviors and societal outcomes took on a newfound urgency in 2020, as governments struggled to limit the spread of COVID-19 (West et al., 2020). Three outcomes that prior research indicates are important for pandemic mitigation are social distancing (McGrail et al., 2020), uptake of vaccines (Huang et al., 2022), and resistance to COVID-19 conspiracy theories which, if

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endorsed, can breed skepticism of health authorities and their recommendations (Farhart et al., 2022). Research has demonstrated that individual differences in personality are important predictors of each of these outcomes (Aschwanden et al., 2021; Enders et al., 2023; Ludeke et al., 2021; Sprengholz et al., 2023).

Much of this work has focused on the role of Big Five personality traits (e.g., Aschwanden et al., 2021; Han, 2021; Ludeke et al., 2021). These traits are Openness to Experience, Conscientiousness, Extraversion, Agreeableness, and Negative Emotionality (John & Srivastava, 1999; Soto & John, 2017). Personality is an important predictor of people's behaviors and life outcomes across many domains (Soto, 2019, 2021). Therefore, the Big Five should also be related to the way that people respond to the COVID-19 pandemic.

Taking these traits in order of their conventional presentation, Openness to Experience corresponds to enthusiastic engagement with novel ideas and aesthetics. One theory predicts that Open individuals are better able to understand or adapt to the gravity of the pandemic and to thereby see the necessity of behavioral restrictions (e.g., Sprengholz et al., 2023). However, prior research indicates that Openness is related to liberal political ideology (McCrae, 1996; Sibley et al., 2012; Vitriol et al., 2019), and political ideology is a robust predictor of COVID-19 behavior and vaccine uptake (Farhart et al., 2022). Thus, Ludeke et al. (2021) note that Open individuals' political affiliations may result in them receiving more messages from media and elites emphasizing the importance of pandemic-mitigation efforts, endorsing vaccination, and discrediting conspiracy theories (see also Motta et al., 2020; Ollerenshaw, 2022). While the mechanism remains unclear, several studies report that Openness is positively related to social distancing (Al-Omiri et al., 2021; Fink et al., 2021; Götz et al., 2021; Ludeke et al., 2021; Xie et al., 2020).

People high in Conscientiousness tend to be orderly and rule-following, attending diligently to tasks—including those related to health and hygiene (Bogg & Roberts, 2004; Hampson & Friedman, 2008; Roberts et al., 2014). Throughout the pandemic, highly Conscientious people were more likely to abide by government recommendations to social distance and get vaccinated (Adamus et al., 2022; Chan et al., 2021; Götz et al., 2021; Ludeke et al., 2021). Conscientiousness has also been linked to the ability to disregard information that is deemed irrelevant to their current goals (Rueter et al., 2018). This tendency may mitigate against conspiratorial thinking, which often involves disordered thought patterns and fixation on irrelevant stimuli (Bowes et al., 2021; Dagnall et al., 2015).

Extraverted individuals tend to exhibit approach-oriented behavior and are motivated to seek out rewarding social interactions. Thus, there is a strong expectation that Extraverts will be more likely to flaunt social distancing guidelines in order to socialize (Cooper et al., 2000) and be less discriminating in their consumption of information (Ahmed & Tan, 2022; Lai et al., 2020), leading to greater endorsement of conspiracy theories and opposition to vaccines. Studies testing these expectations have reported inconsistent results (Aschwanden et al., 2021; Blagov, 2021; Ebrahimi et al., 2021; Shook et al., 2020; Siritzky et al., 2022).

Agreeable people tend to be more pro-social, self-sacrificing, and community oriented, eagerly adopting proscriptions against risky and unhealthy behaviors (Hampson et al., 2007; Turiano et al., 2018). Agreeableness is also related to adherence to government authorities (Greene & Robertson, 2017). Thus, Agreeableness has been observed to correlate with the tendency to comply with health guidelines (Blagov, 2021; Han, 2021; Siritzky et al., 2022). However, because some government officials endorsed conspiracy theories while others rebutted them, Agreeable people's endorsement of conspiracy theories may vary depending on whether this behavior is normative in their local context (Farhart et al., 2022).

Negative Emotionality corresponds to a tendency to experience negative emotions and rapid changes in mood. People high in Negative Emotionality experience heightened anxiety and threat, including in response to disease cues (Duncan et al., 2009). Thus, they might be expected to adhere to social distancing out of fear (Yıldırım et al., 2021). While some studies report a positive relationship between Negative Emotionality and pandemic mitigation-related behaviors (Abdelrahman, 2022; Fink et al., 2021), others find instead a negative relationship (Siritzky et al., 2022; Weismüller et al., 2021) or no relationship at all (Al-Omiri et al., 2021; Pilch et al., 2021).

We contribute to this literature in two ways. First, we attempt to replicate existing findings linking personality traits to pandemic attitudes and behaviors across two independent samples (Sample 1 N = 510; Sample 2 N = 441) collected in October 2020, including a nationally representative sample of Americans. Specifically, we analyze

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the relationship between Big Five personality traits and three critical beliefs and behaviors in the context of the pandemic. These are vaccine hesitancy (VH), engagement in risky (vs. self-protective) pandemic social behaviors, and conspiratorial beliefs surrounding the origins, consequences, and public health respond to COVID-19. Second, we argue that existing accounts that attribute pandemic behaviors to the personality trait Openness to Experience overlook the well documented relationship between Openness and political ideology, rendering these results difficult to interpret. Drawing on theory from political psychology, we model the joint effects of personality and ideology to evaluate whether political liberalism mediates the relationship between Openness and COVID-related attitudes and behaviors.

1.1 | Hypotheses

Thus, we expect that:

- Hypothesis 1: Openness will predict less endorsement of COVID-19 conspiracy theories (hereafter CCTs), less vaccine hesitancy (hereafter VH), and fewer risky social behaviors (hereafter RSBs).
- Hypothesis 2: Conscientiousness will predict less endorsement of CCTs, less VH, and fewer RSBs.
- Hypothesis 3: Agreeableness will predict less VH and fewer RSBs.
- Hypothesis 4: The relationship between Openness and the dependent variables will be mediated by liberal (vs. conservative) ideological self-placement.
- Given the mixed results of prior research, we do not make any predictions for Negative emotionality or Extraversion and treat our analyses of these traits as exploratory.

2 | METHODS

2.1 | Data and participants

We test these hypotheses using data from two panel studies, including one sampled to approximate national representativeness, recruited before (Wave 1) and after the 2020 U.S. Presidential Election (Wave 2). We use data exclusively from Wave 1 of each study. The first sample was recruited from Amazon's Mechanical Turk platform (MTurk). MTurk samples are more diverse than student samples and more representative than typical Internet samples (Berinsky et al., 2012; Mason & Suri, 2012). The second study was quota sampled by Forthright, an online research panel with national representative sampling capabilities, part of market research agency Bovitz, Inc. We use data from Wave 1 of both panel studies. Sample 1 was collected October 23–30, 2020 (N = 1080, 60.4% female and 39.5% male, 56.3% with at least a bachelor's degree (BA), 54.8% with family income greater than $50,000, mean age = 44.18, SD = 14.66; White = 83.98%); Sample 2 was collected October 27–November 2, 2020 (N = 1140, 52.80% female and 46.50% male, 32.6% with at least a BA, 50.7% with family income greater than $50,000, mean age = 46.34, SD = 17.04; White = 62.11%). In both samples, due to limited space in the survey, we utilized a split ballot technique, in which participants were randomly assigned to complete a subset of the measures. We rely upon data from participants who completed *all* measures used in the analysis described below (Sample 1 N = 506; Sample 2 N = 441). Supporting Information S1 has additional info about the demographic characteristics of the sample included in the analysis below.

2.2 | Measures

Means (SD), alphas, and intercorrelations between key variables are shown in Table 1. Question wording and response scales are available in Supporting Information S2. All measures were rescaled to range from 0 to 1 to aid interpretation of

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TABLE 1 Means (*SD*), alphas, and intercorrelations.

	M (*SD*)	Alpha	1	2	3	4	5	6	7	8	9
Sample 1											
1. Openness	0.69 (0.21)	.81	--								
2. Conscientiousness	0.71 (0.23)	.71	***0.26	--							
3. Extraversion	0.48 (0.21)	.52	***0.15	***0.26	--						
4. Agreeableness	0.70 (0.20)	.55	***0.29	***0.32	+0.08	--					
5. Negative emotion	0.41 (0.28)	.78	***−0.16	***−0.52	***−0.39	***−0.25	--				
6. Ideology	0.49 (0.32)	--	***−0.26	0.07	*0.13	−0.05	**−0.18	--			
7. COVID-19 CTs	0.26 (0.21)	.82	***−0.22	***−0.19	**0.14	**−0.18	*0.11	***0.37	--		
8. Vaccine hesitancy	0.29 (0.19)	.82	***−0.20	***−0.18	***0.16	***−0.16	0.06	***0.39	***0.65	--	
9. Risky Soc. Behav.	0.16 (0.22)	.78	***−0.23	***−0.17	+−0.08	***−0.25	+0.08	***0.24	***0.30	***0.24	--
Sample 2											
1. Openness	0.66 (0.19)	.73	--								
2. Conscientiousness	0.68 (0.23)	.63	***0.23	--							
3. Extraversion	0.49 (0.22)	.52	***0.23	***0.33	--						
4. Agreeableness	0.70 (0.19)	.45	***0.30	***0.36	*0.10	--					
5. Negative emotion	0.47 (0.26)	.71	+−0.08	***−0.46	***−0.29	***−0.27	--				
6. Ideology	0.48 (0.28)	--	***−0.21	***0.14	***0.13	+−0.09	***−0.20	--			
7. COVID-19 CTs	0.31 (0.20)	.80	*−0.11	0.05	***0.18	*−0.10	0.02	***0.27	--		
8. Vaccine hesitancy	0.32 (0.19)	.79	***−0.16	−0.02	**0.13	*−0.11	0.05	***0.24	***0.58	--	
9. Risky Soc. Behav.	0.19 (0.23)	.77	***−0.23	**−0.14	−0.07	***−0.20	0.05	***0.22	***0.20	***0.21	--

+$p < 0.1$; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

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unstandardized coefficients. To aid interpretation, we report both unstandardized and standardized coefficients. Replication data and syntax are available at: https://osf.io/6mvrt/?view_only=f9e87594f1754698818275496096a2e7.

2.2.1 | Dependent variables

The three dependent variables include Belief in COVID-19 Conspiracy Theories, Risky COVID-19 Social Behaviors, and VH.

Belief in COVID-19 Conspiracy Theories (CCTs). Our first dependent variable is an index of CCTs, consisting of the average of responses to eight CT beliefs. Prior work indicates that CCTs may comprise part of a coherent belief system (Miller, 2020), and so the use of a composite index is justified. Higher values correspond to increased support for COVID-19 CTs.

Vaccine Hesitancy (VH). Scholars have utilized multiple measures of VH, ranging from a single item assessing vaccine intention (Funk & Tyson, 2020) to more varied measures assessing multiple dimensions of vaccination (Betsch et al., 2018). Our dependent variable, based on Betsch et al. (2018) is a measure of vaccine support and hesitancy. This measure is an index of five items scored on a one to seven agreement scale. Respondents were asked items such as, "I am completely confident that vaccines are safe," and "When everyone else is vaccinated, I don't have to be vaccinated, too." The five items were averaged. Higher values correspond to greater VH and less vaccine support; whereas lower values correspond to less VH and more vaccine support.

Risky COVID-19 Social Behaviors (RSBs). Participants were asked whether they engaged in 8 different behaviors to "prevent the spread of coronavirus" (Yes, No, I Don't Know), including "changed travel plans," "worked from home," "regularly washed hands," "avoided dining-in at restaurants," "canceled social engagements," "sanitized your home or workspace with disinfectant," "stayed at least six feet away from others in public places," and "wore a mask in public." These items were developed specifically for this study. We coded each "Yes" response a 1 and "No" or "I don't know" responses as a 0, and then averaged across eight responses to create our primary outcome measure, a composite measure of RSBs.[1] Equivalent self-report measures have accrued recent empirical validation with objectively assessed behaviors (Gollwitzer et al., 2020). Higher values correspond to decreased RSBs and more protective behaviors in the context of the COVID-19 pandemic.

2.2.2 | Independent variables

Big Five Personality Traits. Personality traits were assessed with the Big Five Inventory 2-XS (BFI-2-XS; Soto & John, 2017), which includes at least one pro-trait and at least one con-trait item (reverse-coded) for each Big Five domain, which were averaged to form an index of each trait. Responses were provided on a five-point Likert-type scale, ranging from "Disagree strongly" to "Agree strongly."

2.2.3 | Mediating variable

In some of our analyses, we use *ideological self-placement* as a mediating variable. This was assessed on a seven-point Likert-type scale (ranging from *Very Liberal*, *Liberal*, *Slightly Liberal*, *Moderate*; *Slightly Conservative*, *Conservative*, *Very Conservative*). This measure of ideology is widely used in both political science and psychology and has been included in the General Social Survey and American National Election Studies since 1972 (Kinder & Kalmoe, 2017). Higher scores correspond with higher levels of political conservatism.

2.2.4 | Control variables

We control for demographic variables, including *age*, *income* (ordered categories rescaled from 0 to 1), *gender* (0 = male, 1 = female), *education* (ordered categories rescaled from 0 to 1), and *race* (0 = nonwhite, 1 = white).

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3 | RESULTS

Bivariate correlations among key variables are available in Table 1. Overall, the pattern of bivariate relationships indicates that Openness and Agreeableness covary with less support for CCTs, fewer RSBs, and less VH. Conscientiousness shows a similar pattern of relationships but is unrelated to support for CCTs and VH in Sample 2. In contrast, Negative Emotionality is largely unrelated to the outcome variables. Finally, Extraversion correlates with more VH and support for CCTs, but is unrelated to RSBs.

3.1 | Main effects of Big 5 personality trait on COVID-19 beliefs and behaviors

First, we examined the main effects of Big Five traits across our three dependent variables. We estimated a series of ordinary least squares regression models. In each sample, we regressed each dependent variable on all independent variables and covariates. To guard against heteroscedasticity, standard errors and confidence intervals were computed using HC3 variance estimates (Long & Ervin, 2000). For this analysis, we include demographic variables as covariates. Given the 0–1 coding of all variables, the coefficients represent the proportion change (or percentage change when multiplied by 100) in the dependent variable associated with going from the lowest to the highest level of each predictor. The results of this analysis for Sample 1 and 2 are summarized in Table A1 in Supporting Information S3 and graphically represented in Figure 1.

COVID-19 CTs. We observed that increased support for CCTs consistently covaried with higher levels of Extraversion (Sample 1, $b = 0.22$, $\beta = .23$, $p < 0.001$; Sample 2, $b = 0.21$, $\beta = .23$, $p < 0.001$) and lower levels of Openness (Sample 1, $b = -0.19$, $\beta = -.15$, $p < 0.001$; Sample 2, $b = -0.15$, $\beta = -.15$, $p = 0.002$). We also observed a negative relationship with Conscientiousness in Sample 1 ($b = -0.10$, $\beta = -.11$, $p = 0.029$) but not Sample 2 ($b = 0.08$, $\beta = .09$, $p = 0.11$). These results align with our expectations for Openness and, to a lesser extent, Conscientiousness. They

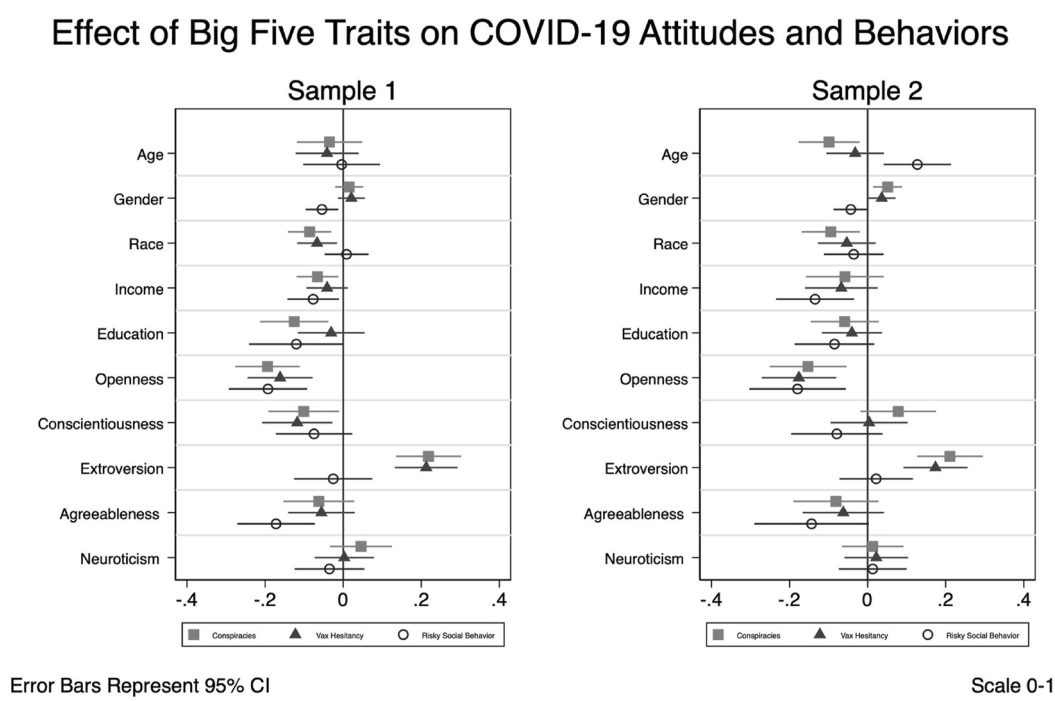

FIGURE 1 Multiple regression results. Estimates represent unstandardized coefficients.

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also align with theories that link extraversion to CT endorsement which have been only inconsistently supported by previous studies.[2]

Vaccine Hesitancy. We observed that increased VH (vs. support) consistently covaried with lower levels of Openness (Sample 1, $b = -0.16$, $\beta = -.18$, $p < 0.001$; Sample 2, $b = -0.18$, $\beta = -.18$, $p < 0.001$) and higher levels of Extraversion (Sample 1, $b = 0.21$, $\beta = .23$, $p < 0.001$; Sample 2, $b = 0.17$, $\beta = .20$, $p < 0.001$). We again observed a negative relationship with Conscientiousness in Sample 1 ($b = -0.12$, $\beta = -.14$, $p = 0.029$) but not Sample 2 ($b = 0.00$, $\beta = .01$, $p = 0.94$). These results align with our expectations for Openness, and, to a lesser extent, Conscientiousness. However, we did not observe a relationship for Agreeableness.

Risky Social Behaviors. We observed that engaging in fewer RSBs was consistently associated with higher levels of Openness (Sample 1, $b = -0.19$, $\beta = -.15$, $p < 0.001$; Sample 2, $b = -0.18$, $\beta = -.15$, $p = 0.004$). Furthermore, engaging in fewer RSBs was significantly associated with Agreeableness in Sample 1 ($b = -0.17$, $\beta = -.15$, $p = 0.001$) but not Sample 2 ($b = -0.15$, $\beta = -.12$, $p = 0.055$). These results align with our expectations for Openness and, to a lesser extent, Agreeableness. In contrast, we fail to observe the hypothesized negative relationship between RSBs and Conscientiousness.

3.2 | Does ideology mediate the effects of Big 5 traits on COVID-19 beliefs and behaviors?

Across all DVs, Openness was a consistent and robust predictor of our dependent variables. Therefore, we proceed to examine the indirect effects of Openness through ideological self-placement. To test this, we use the bootstrap-based method recommended by Preacher and Hayes (2004), in which 5000 bootstrap-replications were used to estimate confidence intervals. Bootstrapping estimates the indirect effect on each resampled data set based on the null hypothesis that the indirect effect is not different from zero. We reject the null hypothesis if the confidence interval does not include zero (Preacher & Hayes, 2004). Political ideology was submitted as the mediator and demographic variables were included as covariates. The results of this analysis are summarized in Tables A2, A3, and A4 in Supporting Information S3.

Across both samples, we observe significant indirect effects of Openness through political ideology on CCTs, RSBs, and VH (and inconsistent direct effects, suggesting much of these relationships are fully mediated). Together, this analysis indicates that the relationship between Openness and attitudes, beliefs, and behaviors about COVID-19 can be explained, in part, by their shared association with political ideology.

4 | DISCUSSION

We analyzed two samples collected during the first year of the COVID-19 pandemic, including one representative sample of US adults. Consistent with previous work and our Hypothesis 1, we found that people higher in Openness were less likely to endorse CCTs, reported lower VH, and engaged in fewer RSBs. Our tests of Hypothesis 2 yielded mixed results. We found the predicted negative relationship between Conscientiousness and both CCTs and VH, but only in Study 1. Given that Study 2, but not Study 1, was representative of the US population, it may be that this relationship holds only among the younger, more liberal MTurk workers sampled in Study 1 (Huff & Tingley, 2015). However, the lack of a relationship between Conscientiousness and RSBs is puzzling in light of existing evidence that supports such an association (Chan et al., 2021; Ludeke et al., 2021). Our tests of Hypothesis 3 also yielded mixed results. We found the expected negative relationship between Agreeableness and RSBs in study 1, but we did not find the hypothesized negative relationship between Agreeableness and VH. This null effect may be because Agreeableness does not guard against CTs, which are a major predictor of VH (Farhart et al., 2022). In contrast to previous research that finds mixed results for Extraversion, we found that the trait was a robust predictor of CCTs

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and VH (Aschwanden et al., 2021; Blagov, 2021). Though unexpected, these findings are consistent with theoretical expectations that link Extraversion to less discriminating information consumption (Ahmed & Tan, 2022; Lai et al., 2020). Finally, consistent with previous research that finds mixed results, we find no relationship between Negative Emotionality and any of our outcome variables (Abdelrahman, 2022; Weismüller et al., 2021).

Across both of our samples, we found strong support for the indirect effect of Openness through political ideology on COVID-related behaviors as predicted by Hypothesis 4. These results demonstrate that Openness does not necessarily contribute directly to pandemic-related attitudes and behaviors as some previous studies have suggested (e.g. Sprengholz et al., 2023). Rather, highly Open individuals may be more likely to adopt attitudes and engage in behaviors geared towards mitigating the spread of COVID-19 because they are more likely to receive and accept appeals from information sources associated with the political left (Farhart et al., 2022). In contrast to conservative media, these information sources were far less likely to downplay COVID-19 or endorse CTs (Motta et al., 2020). Our findings regarding Openness are consistent with research in political psychology that emphasizes the importance of information environments in generating relationships between personality and attitudes (Federico & Goren, 2009). Future research would benefit from pursuing similar analyses with other traits.

These results must be considered with the limitations of the present study in mind. The brevity of our measures of personality and ideology will have attenuated the observed relationships among study variables, potentially accounting for some unexpected null results. Additionally, our use of cross-sectional data for mediation tests is not ideal—if also representing a limitation that is difficult to meaningfully escape for the present study question. Future research might explore the topic with panel data from a broader range of national contexts.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to report.

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ENDNOTES

[1] Neither the results nor our interpretation change meaningfully if we omit "I don't know" responses from our analysis.

[2] Our measure of belief in COVID-19 Conspiracy Theories includes two items that mention Donald Trump. We found that excluding or including these items did not meaningfully change any of the relationships between our CCTs variable and our predictor variables.

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